RAYHAN MUHAMMAD ALFARIZI

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Electrical Engineering graduate with a concentration in Information Technology and hands-on experience in full-stack and backend development, machine learning, and IoT systems. Experienced in building RESTful APIs, data-driven applications, and real-time IoT services, with a strong foundation in software engineering, databases, Git, and modern web development. Particularly interested in scalable backend and cloud-native systems.

EDUCATION

Diponegoro University | Bachelor of Electrical Engineering **2022 – 2026**

- Information Technology Concentration | GPA: 3.87 / 4.00
- Relevant Coursework: Algorithms and Programming, Data Structures, Object-Oriented Programming, Software Engineering, Database, Operating Systems, Web Development, Modern Programming Methods

EXPERIENCE

Teaching Assistant – Algorithms and Programming **Feb – Jun 2025**
Diponegoro University

- Assisted 30 students in **Python** and **C++** programming laboratory sessions.
- Guided programming fundamentals and debugging.

Web Development Intern **Jan – Feb 2025**
PT Des Teknologi Informasi (DESNET)

- Developed backend features for a web-based Guest Management System using **CodeIgniter 4**, including APIs for guest form submissions and application logic for the dashboard, visit history, and scheduling.
- Implemented user authentication and role-based authorization and contributed **JavaScript** functionality to support scheduling interface interactivity.
- Collaborated with a two-person development team using **Git and GitHub**, participating in code reviews, task planning, and feature integration.

PROJECTS

Autoencoder-Based Fault Detection & Diagnosis for Smart Precision Irrigation **Jan – Jul 2026**

- Built an end-to-end IoT monitoring and fault diagnosis platform using **ESP32 and MQTT** for sensor data communication, **FastAPI** for backend processing, and **Keras** for Autoencoder-based anomaly detection, achieving **97.56% F1-score**.
- Used **InfluxDB** for time-series data, **MariaDB** for application data, and **Laravel with Livewire** for the monitoring dashboard; containerized and deployed services using **Docker** on a VPS.

Artikula: Reading Space **Feb 2026**

- Developed a full-stack article publishing platform using **Laravel and Livewire** for application logic and interactive interfaces, with **Supabase PostgreSQL** for relational data management.
- Implemented authentication, article management, search, filtering, analytics, and feedback features.

Bubble Pop Game **Nov – Dec 2025**

- Built an interactive browser game using **React and TypeScript** for UI and game logic, and **Tailwind CSS** for responsive styling.
- Implemented real-time scoring, animations, pause functionality, and customizable difficulty settings.

Letsgobois Shop – E-Commerce Platform **Sep – Dec 2025**

- Developed an e-commerce platform using **Next.js**, with **Firebase and Google Authentication** for user management and **Midtrans** for online payment processing.

Movie App **Aug 2025**

- Built a movie discovery application using **React**, integrated the **TMDB API** for movie data, and used **Tailwind CSS** for the responsive interface.
- Implemented search, category filtering, movie details, and watchlist functionality.

SI PANDA – Sistem Pantau Data Anak **Jul – Aug 2025**

- Developed a child growth monitoring application using **CodeIgniter 4** for application logic, **Supabase PostgreSQL** for database, and **Tailwind CSS** for styling.
- Implemented child data management, measurement records, search, data visualization, and monthly recaps.

SI JEMPANG – Sistem Jentik Pantau Kampung **Jul – Aug 2025**

- Developed a mosquito larvae monitoring application using **CodeIgniter 4** and **Supabase** for data management, with **Leaflet.js** for location-based visualization and monitoring.

PUBLICATIONS

- Autoencoder-Based Fault Detection and Diagnosis for IoT-Enabled Smart Precision Irrigation of Avocado Trees | Accepted for presentation at IEEE ICITACEE Conference 2026.

LEADERSHIP

Director of Event – Elektro Forge (E-Forge) **Aug – Sep 2024**

Electrical Engineering Student Association (HME), Diponegoro University

- Led a 5-member organizing committee delivering a PCB Design Training for 54 participants.
- Coordinated planning, logistics, and stakeholder communication to ensure successful event execution.

CERTIFICATIONS

- Python Essentials 2 – Cisco Networking Academy
- Python Essentials 1 – Cisco Networking Academy
- C Essentials 1 – Cisco Networking Academy
- Machine Learning for Beginners — Dicoding Indonesia
- Data Analysis with Python — Dicoding Indonesia
- Basic SQL — Dicoding Indonesia

LANGUAGES

- Indonesian – Native
- English – Professional Working Proficiency
  SEU TOEFL Prediction Test (Equivalent Score: 513)